CS464 Homework 1: IMDb Movie Review Classification

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Question 1.1: Multinomial Naive Bayes without Smoothing

Data and class distribution. I used the supplied bag-of-words features with a vocabulary of $V = 2000$ words and preserved the training/validation split. The data is a teaching subset of the IMDb dataset [1]. The class counts and percentages within each partition are:

Partition	Negative (0)	Positive (1)	Total
Training	800 (50.00%)	800 (50.00%)	1600
Validation	200 (50.00%)	200 (50.00%)	400

The training priors are therefore

$$\pi_0 = \frac{N_0}{N} = \frac{800}{1600} = 0.5, \qquad \pi_1 = \frac{N_1}{N} = \frac{800}{1600} = 0.5.$$

The training set is balanced, so the prior terms do not favor either class. With class imbalance, the majority class would have a larger estimated prior. Its larger log prior would favor that class when the word evidence is similar, although sufficiently strong evidence could still favor the minority class.

Parameter estimation and prediction. For class k, I summed word occurrences over its training reviews, including repeated occurrences:

$$T_{jk} = \sum_{i:y_i=k} x_{ij}, \qquad T_k = \sum_{j=1}^{V} T_{jk}, \qquad \theta_{jk} = \frac{T_{jk}}{T_k}.$$

The training token totals were $T_0 = 146358$ and $T_1 = 148742$. I computed $s_k(\mathbf{x}_i) = \log \pi_k + \sum_j x_{ij} \log \theta_{jk}$ using natural logarithms and predicted 1 only if $s_1 > s_0$. Every tie, including two $-\infty$ scores, predicts 0. An absent word contributes zero even when its probability is zero. If an observed word has zero probability for a class, I explicitly set that class score to $-\infty$, avoiding `NaN` values without changing the estimated probabilities. All parameters were estimated from training data only.

Validation results. Throughout this report, confusion matrix rows are true labels and columns are predicted labels, ordered negative (0), positive (1).

$$C_{\text{unsmoothed}} = \begin{array}{c|cc} & \widehat{y}=0 & \widehat{y}=1 \\ \hline y=0 & 171 & 29 \\ y=1 & 42 & 158 \end{array}$$

$$\text{Accuracy} = \frac{171+158}{400} = \boxed{0.8225}, \qquad \text{Incorrect} = 29+42 = \boxed{71}.$$

There were **5** vocabulary words with zero estimated probability in the negative class and **7** in the positive class.

Question 1.2: Laplace Smoothing

I used additive smoothing with $\alpha = 1$, keeping the same training priors, vocabulary, features, and validation set:

$$\theta_{jk}^{(\alpha)} = \frac{T_{jk} + \alpha}{T_k + \alpha V}.$$

For the two classes, this gives

$$\theta_{j0}^{(1)} = \frac{T_{j0} + 1}{146358 + 2000} = \frac{T_{j0} + 1}{148358}, \qquad \theta_{j1}^{(1)} = \frac{T_{j1} + 1}{148742 + 2000} = \frac{T_{j1} + 1}{150742}.$$

The priors remain $\pi_0 = \pi_1 = 0.5$. I used the same natural-log scoring rule and the same tie convention as in Question 1.1.

Why all word probabilities are positive. Every count satisfies $T_{jk} \geq 0$, so adding one makes every numerator at least one. Each denominator is positive. Thus every word in the fixed vocabulary has positive probability in both classes, including words never observed in one of the classes. The probabilities still sum to one:

$$\sum_{j=1}^{V} \theta_{jk}^{(1)} = \frac{\sum_j T_{jk} + V}{T_k + V} = 1.$$

The numbers of zero-probability words consequently become **0** for the negative class and **0** for the positive class. This guarantee applies to the supplied vocabulary; words outside it are not represented in the feature vectors.

Validation results. The smoothed model produced the following confusion matrix:

$$C_{\text{Laplace}} = \begin{array}{c|cc} & \widehat{y} = 0 & \widehat{y} = 1 \\ \hline y = 0 & 171 & 29 \\ y = 1 & 43 & 157 \end{array}.$$

Therefore,

$$\text{Accuracy} = \frac{171 + 157}{400} = \boxed{0.8200}, \qquad \text{Incorrect} = 29 + 43 = \boxed{72}.$$

Comparison of measured results.

Model	Accuracy	Incorrect	Zero words (0)	Zero words (1)
Unsmoothed	0.8225	71	5	7
Laplace ($\alpha = 1$)	0.8200	72	0	0

Laplace smoothing removes zero probabilities but gives one more incorrect prediction overall on this validation set. The numbers in both tables were computed from the supplied CSV files by `main.py`.

Question 1.3: Discussion

1. Accuracy comparison and the effect of smoothing

The unsmoothed model has higher validation accuracy. The difference is

$$(0.8225 - 0.8200) \times 100 = \boxed{0.25 \text{ percentage points}}.$$

Smoothing does not necessarily improve validation accuracy. It prevents zero word probabilities, but it also changes the probabilities of every word in each class. In fact, for $\alpha > 0$ the estimator can be written as

$$\theta_{jk}^{(\alpha)} = \frac{T_k}{T_k + \alpha V}\theta_{jk} + \frac{\alpha V}{T_k + \alpha V}\frac{1}{V}.$$

Thus smoothing moves the empirical word distribution toward the uniform distribution. This can reduce the effect of sparse counts, but it may also weaken useful differences between the classes and change some decisions adversely. Here, $\alpha = 1$ produces a net increase of one validation error. This small difference on one split does not establish that the unsmoothed model will always perform better on other data.

2. Words unseen in one training class

If an observed word has never occurred in training reviews of class k, the unsmoothed estimate is $\theta_{jk} = 0$. Its positive count in the review makes $s_k = -\infty$, regardless of the other words. If the other class has a finite score, it is selected. If both class scores are $-\infty$, the specified tie rule selects label 0. A zero-probability word that is absent from the review has no effect.

With Laplace smoothing, that word receives probability $1/(T_k + V)$, which is positive. Its log contribution is finite, so the class is no longer ruled out by that word alone. The prediction instead depends on the combined contributions of all observed words and the prior.

3. Class-specific errors and the limits of accuracy

The unsmoothed confusion matrix shows **29** errors among true negative reviews and **42** among true positive reviews. These are error rates of $29/200 = 14.50\%$ and $42/200 = 21.00\%$, respectively. For Laplace smoothing, the corresponding error counts are **29** and **43**, giving rates of 14.50% and 21.50%. Thus the **positive class has more incorrect predictions in both models**. Since the validation classes have equal size, it also has the higher class-specific error rate in both cases.

On a strongly imbalanced dataset, overall accuracy can conceal poor minority-class performance. For example, if 95% of reviews are negative, always predicting negative gives 95% accuracy while missing every positive review. Confusion matrices and per-class recall reveal this failure; balanced accuracy and macro-averaged scores can also help evaluate performance across classes.

Acknowledgement

I used ChatGPT (GPT-6) to improve the language used throughout the document and correct grammatical errors.

References

[1] Andrew L. Maas, Raymond E. Daly, Peter T. Pham, Dan Huang, Andrew Y. Ng, and Christopher Potts. Learning Word Vectors for Sentiment Analysis. In *Proceedings of ACL*, pp. 142–150, 2011. `https://aclanthology.org/P11-1015/`.